Aayush Bathija

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Education

Oak Park High School Oak Park, CA
Class of 2029

Moorpark College, dual enrollment 2022–present
Calculus I, II. Psychology, Symbolic Logic, Philosophy, and General Physics I

Independent study. Real analysis through Tao's *Analysis I.*

Papers

- A. Bathija, P. Rohatgi, D. Soskin. *Bounded Ratios for Lorentzian Polynomials.* arXiv:2609.05341, 2026.

Research

Bounded ratios for Lorentzian polynomials 2026–present
Extending the Huang–Huh–Soskin–Wang results to general degree: a dual description of the bounded-ratio cone in any degree, optimal constants for ternary Lorentzian cubics, and a classification of when the Lorentzian and linear forms Polynomial Coincide. With Prince Rohatgi and Prof. Daniel Soskin (UCLA). Funded by ORMC's Vertical Research Integration program.

Ahlswede–Daykin for determinantal inequalities 2026–present
Applying the Ahlswede–Daykin four functions theorem to products of minors of totally positive matrices, yielding interval inequalities on the sorting lattice. Manuscript in preparation with Prince Rohatgi and Prof. Daniel Soskin (UCLA). Funded by ORMC's Vertical Research Integration program.

Fairness of voting systems 2026–present
Simulating ranked-choice, STAR, and related methods and comparing how each one fails. Started in the Non-Trivial fellowship and ongoing.

Teaching

Instructor, UCLA Olga Radko Math Circle 2026–present
Teaching weekly sessions, and taking pure math classes.

Instructor, Agoura Math Circle 2024–2026
Taught and graded weekly classes. Primary on the documentation team, and wrote scripts that migrate documentation.

Volunteer coach, Medea Creek Middle School 2025–present
Running practice for the school's MATHCOUNTS team.

Leadership

- President, Philosophy Club, Oak Park High School. Preparing presentations, organizing talks, and facilitating discussion.
- Founder, Higher Math Club, Oak Park High School. Proof-based material for students who have exhausted competition math.

Awards

- Guinness World Record, most countries identified by outline in one minute
- John Locke Institute Essay Competition, 2x shortlist
- Superintendent's Compassionate and Creative Citizens Award, given to four students district-wide and chosen by the superintendent
- Great History Challenge, national runner-up
- Berkeley Math Tournament, top ten in the Power Round (team)
- AMC 10, Distinction in 2026 (AIME qualifier)

Programs

Pi Math, Group Theory 2026–present
Proof-based online course in group theory with Prof. Paquin.

AwesomeMath Academy 2024–present
Number Theory with Mirroslav Yotov, into algebraic number theory: Gaussian integers, unique factorization, quadratic residues. Also Computational Geometry and Counting Strategies.

Non-Trivial Research Fellowship 2025
Voting systems project

LEAF Program, *The Mathematics of Morality* 2024

Other

Academic Learning (youtube.com/@Academic-Learning), discussing and working through mathematical material.

Speech and debate. 2nd place, Spontaneous Argumentation (2026). Double octofinalist, open Lincoln–Douglas, Jack Howe Memorial Tournament, CSU Long Beach (2026). Reached the final round of the state qualifier.

Speedcubing: 6.58 s single, 3×3.